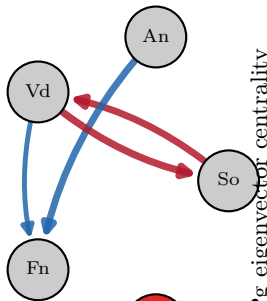
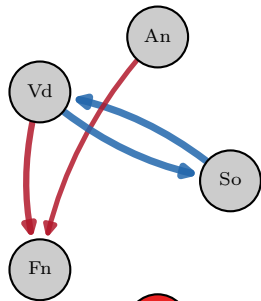


Commensal  $\rightarrow$  dysbiotic network rewiring (Heine 5-species, gauge-invariant  $A_{\text{eff}}$ )  
*Streptococcus-Veillonella* mutualism  $\rightarrow$  competition in dysbiosis  
 Commensal (CS, CH)      Dysbiotic (DS, DH)      Pg network role rises into dysbiosis



— facilitation  
 — competition

